

Date: 2026-08-04
Request ID: 100048838

Hotplate		
Analyte(s)	Ag	
SOP#	Determination of % Silver Composition	
Cocktail		
sample(s)	weight (g)	volume (mL)
200128752_1	0.5057	200
200128752_2	0.5031	200
200128753_1	0.5071	200
200128753_2	0.5010	200
200128754_1	0.5024	200
200128754_2	0.5083	200

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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128752	16.4841	18.4966	18.4847	0.59	0.9941

Ag ICP analysis

sample	Dilution	conc. [mg/L]	Ag_ppm [dried]	%Ag [dried]
200128752_1	10/9	1.784	788.61	0.08
200128752_2	10/9	1.728	767.81	0.08

ag result: 778.21 ppm

ag result: 0.08 %

ag lot: *Manufacturer: Assurance, 28-111AGY exp: 2027-02-28*

Note(s):

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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128753	19.0623	21.0827	21.0712	0.57	0.9943

Ag ICP analysis

sample	Dilution	conc. [mg/L]	Ag_ppm [dried]	%Ag [dried]
200128753_1	10/9	1.715	755.85	0.08
200128753_2	10/9	1.856	827.96	0.08

ag result: 791.90 ppm

ag result: 0.08 %

ag lot: *Manufacturer: Assurance, 28-111AGY exp: 2027-02-28*

Note(s):

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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128754	18.0218	20.0290	19.9898	1.95	0.9805

Ag ICP analysis

sample	Dilution	conc. [mg/L]	Ag_ppm [dried]	%Ag [dried]
200128754_1	10/9	1.754	791.29	0.08
200128754_2	10/9	1.885	840.51	0.08

ag result: 815.90 ppm

ag result: 0.08 %

ag lot: *Manufacturer: Assurance, 28-111AGY exp: 2027-02-28*

Note(s):

$$\text{ppm } M+ = [\text{conc.}][\text{volume}][\text{dilution}]/[\text{weight}]$$

$$\%M+ = [\text{conc.}][\text{volume}][\text{dilution}]/([\text{weight}]*10,000)$$

$$\%MO = [\text{oxide factor}]*\%M+$$

A = crucible

B = crucible + sample

C = crucible + sample after drying

$$\%LOI = ([B-A]-[C-A])/(B-A)*100\%$$

$$\text{ppm } M+ \text{ [dried]} = \text{ppm } M+ / ([1-(\%LOI)/100])$$

$$\%M+ \text{ [dried]} = \%M+ / ([1-(\%LOI)/100])$$

$$\%MO \text{ [dried]} = \%MO / ([1-(\%LOI)/100])$$